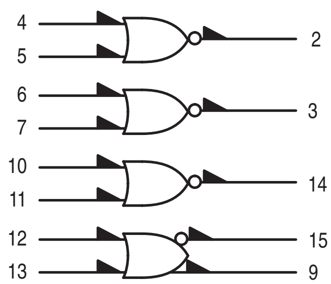
MC10102

Quad 2-Input NOR Gate

The MC10102 is a quad 2–input NOR gate. The MC10102 provides one gate with OR/NOR outputs.

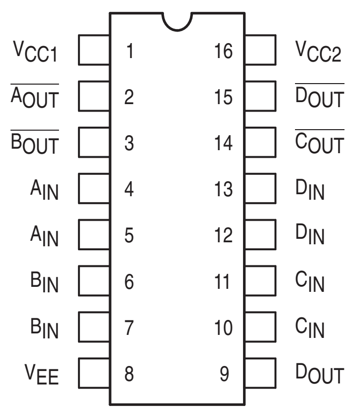
- P_D = 25 mW typ/gate (No Load)
- t_{pd} = 2.0 ns typ
- t_r, t_f = 2.0 ns typ (20%–80%)

LOGIC DIAGRAM



V_{CC1} = PIN 1
V_{CC2} = PIN 16
V_{EE} = PIN 8

DIP
PIN ASSIGNMENT

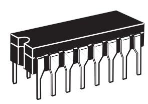


Pin assignment is for Dual–in–Line Package.
For PLCC pin assignment, see the Pin Conversion Tables on page 18.

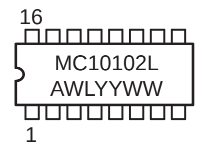
ON Semiconductor

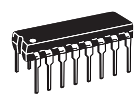
http://onsemi.com

MARKING DIAGRAMS

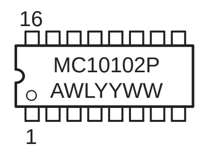


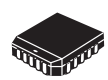
CDIP–16
L SUFFIX
CASE 620



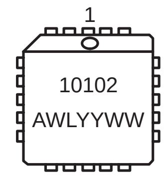


PDIP–16
P SUFFIX
CASE 648





PLCC–20
FN SUFFIX
CASE 775



A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week

ORDERING INFORMATION

Device	Package	Shipping
MC10102L	CDIP–16	25 Units / Rail
MC10102P	PDIP–16	25 Units / Rail
MC10102FN	PLCC–20	46 Units / Rail

ELECTRICAL CHARACTERISTICS

Characteristic		Symbol	Pin Under Test	Test Limits							Unit
				–30°C		+25°C			+85°C		
				Min	Max	Min	Typ	Max	Min	Max	
Power Supply Drain Current		I_E	8		29		20	26		29	mAdc
Input Current		I_{inH}	12		425			265		265	µAdc
		I_{inL}	12	0.5		0.5			0.3		µAdc
Output Voltage	Logic 1	V_{OH}	9 9 15 15	–1.060 –1.060 –1.060 –1.060	–0.890 –0.890 –0.890 –0.890	–0.960 –0.960 –0.960 –0.960		–0.810 –0.810 –0.810 –0.810	–0.890 –0.890 –0.890 –0.890	–0.700 –0.700 –0.700 –0.700	Vdc
Output Voltage	Logic 0	V_{OL}	9 9 15 15	–1.890 –1.890 –1.890 –1.890	–1.675 –1.675 –1.675 –1.675	–1.850 –1.850 –1.850 –1.850		–1.650 –1.650 –1.650 –1.650	–1.825 –1.825 –1.825 –1.825	–1.615 –1.615 –1.615 –1.615	Vdc
Threshold Voltage	Logic 1	V_{OHA}	9 9 15 15	–1.080 –1.080 –1.080 –1.080		–0.980 –0.980 –0.980 –0.980			–0.910 –0.910 –0.910 –0.910		Vdc
Threshold Voltage	Logic 0	V_{OLA}	9 9 15 15		–1.655 –1.655 –1.655 –1.655			–1.630 –1.630 –1.630 –1.630		–1.595 –1.595 –1.595 –1.595	Vdc
Switching Times (50Ω Load)											ns
Propagation Delay		t_{12+15-} t_{12-15+} t_{12+9+} t_{12-9-}	15 15 9 9	1.0 1.0 1.0 1.0	3.1 3.1 3.1 3.1	1.0 1.0 1.0 1.0	2.0 2.0 2.0 2.0	2.9 2.9 2.9 2.9	1.0 1.0 1.0 1.0	3.3 3.3 3.3 3.3	
Rise Time	(20 to 80%)	t_{15+} t_{9+}	15 9	1.1 1.1	3.6 3.6	1.1 1.1	2.0 2.0	3.3 3.3	1.1 1.1	3.7 3.7	
Fall Time	(20 to 80%)	t_{15-} t_{9-}	15 9	1.1 1.1	3.6 3.6	1.1 1.1	2.0 2.0	3.3 3.3	1.1 1.1	3.7 3.7	

ELECTRICAL CHARACTERISTICS (continued)

			TEST VOLTAGE VALUES (Volts)					
		@ Test Temperature	V_{IHmax}	V_{ILmin}	V_{IHAmin}	V_{ILAmax}	V_{EE}	
		–30°C	–0.890	–1.890	–1.205	–1.500	–5.2	
		+25°C	–0.810	–1.850	–1.105	–1.475	–5.2	
		+85°C	–0.700	–1.825	–1.035	–1.440	–5.2	
			TEST VOLTAGE APPLIED TO PINS LISTED BELOW					
Characteristic	**Symbol**	**Pin Under Test**	V_{IHmax}	V_{ILmin}	V_{IHAmin}	V_{ILAmax}	V_{EE}	**(V_{CC}) Gnd**
Power Supply Drain Current	I_E	8					8	1, 16
Input Current	I_{inH}	12	12				8	1, 16
	I_{inL}	12		12			8	1, 16
Output Voltage Logic 1	V_{OH}	9	12				8	1, 16
		9	13				8	1, 16
		15					8	1, 16
		15					8	1, 16
Output Voltage Logic 0	V_{OL}	9					8	1, 16
		9					8	1, 16
		15	12				8	1, 16
		15	13				8	1, 16
Threshold Voltage Logic 1	V_{OHA}	9			12		8	1, 16
		9			13		8	1, 16
		15				12	8	1, 16
		15				13	8	1, 16
Threshold Voltage Logic 0	V_{OLA}	9				12	8	1, 16
		9				13	8	1, 16
		15			12		8	1, 16
		15			13		8	1, 16
Switching Times (50Ω Load)					**Pulse In**	**Pulse Out**	**–3.2 V**	**+2.0 V**
Propagation Delay	t_{12+15-}	15			12	15	8	1, 16
	t_{12-15+}	15			12	15	8	1, 16
	t_{12+9+}	9			12	9	8	1, 16
	t_{12-9-}	9			12	9	8	1, 16
Rise Time (20 to 80%)	t_{15+}	15			12	15	8	1, 16
	t_{9+}	9			12	9	8	1, 16
Fall Time (20 to 80%)	t_{15-}	15			12	15	8	1, 16
	t_{9-}	9			12	9	8	1, 16

Each MECL 10,000 series circuit has been designed to meet the dc specifications shown in the test table, after thermal equilibrium has been established. The circuit is in a test socket or mounted on a printed circuit board and transverse air flow greater than 500 linear fpm is maintained. Outputs are terminated through a 50-ohm resistor to –2.0 volts. Test procedures are shown for only one gate. The other gates are tested in the same manner.